

FINAL INSIGHTS

Project Title: SALES AND PROFIT ANALYSIS

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Dataset: Ecommerce retail dataset. In this dataset we have total Four table:

1) Customer Table

2) Product Table

3) Sales Table

4) Store Table

Key fields:

Store Name, Product Name, Brand Name, First Name, Lastname, Age, Gender, City, State, Country, Loyalty Level, Payment type, Month, Category, Sub Category, Cost, Stock, Region, Store type, Total Amount, Discounted Amount, Profit Margin.

Data cleaning & transformation:

Using Power query

1.Handling Missing Values:

CLEAN FUNCTION: Removes line breaks, charterers and other hidden control.

TRIM FUNCTION: This removes non printable characters and extra spaces at the same time.

PROPER FUNCTION: The text can replace capitalized each word.

ISBLANK FUNCTION: Checks whether cell is truly empty and returns True and False.

Syntax: =IF(ISBLANK(H28),AVERAGE(Sales_Fact[Discount]),H28)

Screenshot of Discount Table:

H
Discount
0.09
0.27
0.25
0.16
0.18
0.19
0.15
0.23
0.07
0.23
0.07
0.17
0.01
0.30
0.29
0.04
0.25
0.23
0.25
0.04
0.27
0.22
0.03
0.08
0.07
0.14
0.15
0.17
0.11

	Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Discounted_Revenue
1	SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	\$218	0.09	Credit Card	\$218	\$198
2	SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	\$104	0.27	PayPal	\$312	\$228
3	SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	\$914	0.25	Credit Card	\$3,656	\$2,742
4	SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	\$922	0.16	COD	\$922	\$774
5	SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	\$745	0.18	Credit Card	\$2,235	\$1,833
6	SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	\$147	0.19	COD	\$441	\$357
7	SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	\$785	0.15	PayPal	\$1,570	\$1,335
8	SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	\$342	0.23	PayPal	\$1,368	\$1,053
9	SALE8	04-05-2025	CUST1187	PROD99	STORE5	2	\$448	0.07	PayPal	\$896	\$833
10	SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	\$873	0.23	PayPal	\$1,746	\$1,344
11	SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	\$279	0.07	Credit Card	\$279	\$259
12	SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	\$287	0.17	Credit Card	\$1,148	\$953
13	SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	\$645	0.01	Credit Card	\$2,580	\$2,554
14	SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	\$317	0.30	PayPal	\$634	\$444
15	SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	\$403	0.29	PayPal	\$1,612	\$1,145
16	SALE15	10-12-2024	CUST1435	PROD36	STORE9	2	\$871	0.04	COD	\$1,742	\$1,672
17	SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	\$452	0.25	Credit Card	\$904	\$678
18	SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	\$503	0.23	Credit Card	\$1,006	\$775
19	SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	\$559	0.25	COD	\$1,118	\$839
20	SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	\$514	0.04	COD	\$2,056	\$1,974
21	SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	\$986	0.27	COD	\$2,958	\$2,159
22	SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	\$590	0.22	Credit Card	\$590	\$460
23	SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	\$979	0.03	PayPal	\$2,937	\$2,849
24	SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	\$719	0.08	Credit Card	\$719	\$661
25	SALE24	08-09-2025	CUST1186	PROD67	STORE6	3	\$78	0.07	Credit Card	\$234	\$218
26	SALE25	17-09-2024	CUST1381	PROD5	STORE3	4	\$853	0.14	PayPal	\$3,412	\$2,934
27	SALE26	03-01-2024	CUST1213	PROD98	STORE8	4	\$156	0.15	Credit Card	\$624	\$530

Answer: 0.15

Syntax: =IF(ISBLANK(Sales_Fact[@[Total_Amount]]),AVERAGE(Sales_Fact[Total_Amount]),Sales_Fact[@[Total_Amount]])Total Amount

Screenshot Total_Amount of Table:

Total_Amount
\$218
\$312
\$3,656
\$922
\$2,235
\$441
\$1,570
\$1,368
\$896
\$1,746
\$279

Syntax: =F2*G2*(1-H2)

Screenshot of Discounted_Revenue Table:

Discounted_Revenue
\$198
\$228
\$2,742
\$774
\$1,833
\$357
\$1,335
\$1,053
\$833
\$1,344
\$259
\$953

Syntax:

=IF(ISBLANK(Product_Dim[@Cost],AVERAGE(Product_Dim[Cost],Product_Dim[@Cost])) cost

Screenshot of Cost Table:

Cost
\$112.00
\$343.00
\$254.00
\$365.00
\$306.00
\$261.00
\$313.00
\$219.00
\$439.00
\$496.00

Syntax:

**=IF(ISBLANK(Product_Dim[@Stock]),AVERAGE(Product_Dim[Stock],
Product_Dim[@Stock])) stock**

Screenshot of Stock Table:

G
Stock
\$416
\$181
\$352
\$452
\$487
\$59
\$350
\$170
\$142
\$323
\$308
\$273
\$460
\$453

- In the Customer table I have created column of “**Loyalty_Level**”.

Customer_Dim - Power Query Editor

Table.TransformColumnTypes(#Renamed Columns1,{"Loyalty_Level", type text})

	Customer_ID	First Name	Last Name	Age	Gender	City	State
1	CUST1000	Sheena	Bonilla	55	Male	Donaldton	North Carolina
2	CUST1001	Eugene	Nelson	26	Female	Phillipsmouth	North Dakota
3	CUST1002	Dr.Benjamin	Price	29	Female	Kylebury	Alabama
4	CUST1003	Jeremy	Harris	59	Female	Thomastown	Ohio
5	CUST1004	Caroline	Collins	64	Female	Nicholasshire	California
6	CUST1005	Christine	Vaughn	24	Male	East Amandamouth	Oregon
7	CUST1006	Amber	Jackson	29	Female	Deborahborough	South Dakota
8	CUST1007	Daniel	Boyd	29	Female	West Pamelafurt	West Virginia
9	CUST1008	Melissa	Howard	53	Female	Maynardthon	New Hampshire
10	CUST1009	Ms.Maria	Leach	64	Male	Meliosashire	Pennsylvania
11	CUST1010	Dominic	Page	53	Male	Port Michaelburgh	Pennsylvania
12	CUST1011	Brian	Lopez	28	Female	East Krista	New York
13	CUST1012	Kenneth	Lambert	53	Female	North Kathryn	Kansas
14	CUST1013	Nathan	Hernandez	32	Female	New Becky	Montana
15	CUST1014	Ashley	Cohen	59	Female	Lake Kristamouth	Nebraska
16	CUST1015	Laura	Alvarez	49	Female	Lawsonhaven	Washington
17	CUST1016	John	Scott	24	Female	South Christopher	Pennsylvania
18	CUST1017	Adam	Cruz	60	Female	Brandonville	Georgia
19	CUST1018	Rhonda	Moss	43	Male	South William	Kansas
20	CUST1019	Michael	Torres	46	Female	Gonzalezfort	New Jersey
21	CUST1020	Bridget	Allen	59	Male	Lake Shaneburgh	Delaware
22	CUST1021	Jeff	Martinez	36	Female	Smithburgh	Oklahoma
23	CUST1022	Paul	Fox	33	Female	Michaelton	Hawaii
24	CUST1023	Jonathan	Steele	52	Female	Molinabury	Oregon
25	CUST1024	Heather	Collins	40	Male	Rogerschester	Idaho
26	CUST1025	Diana	Taylor	59	Female	Vasqueztown	California
27	CUST1026	Amber	Carlson	18	Female	East Michellemouth	Missouri
28	CUST1027	Leslie	Johnson	31	Male	Mollyshire	Utah
29	CUST1028	Joshua	Ryan	27	Female	New David	New Jersey
30	CUST1029	David	Arnold	54	Female	East Jennifer	California
31	CUST1030	Jessica	Acevedo	64	Female	Amybury	Pennsylvania
32	CUST1031	Sarah	Vance	41	Male	Johnsonville	Oregon

Query Settings

PROPERTIES

Name: Customer_Dim

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Removed Duplicates

Cleaned Text

Trimmed Text

Replaced Value

Filtered Rows

Cleaned Text1

Trimmed Text1

Capitalized Each Word

Cleaned Text2

Trimmed Text2

Capitalized Each Word1

Cleaned Text3

Trimmed Text3

Capitalized Each Word2

Cleaned Text4

Trimmed Text4

Capitalized Each Word3

Cleaned Text5

Trimmed Text5

Cleaned Text6

Trimmed Text6

Uppercased Text

Replaced Value1

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Loyalty_Level1

Column Name	Operator	Value	Output
If Age	is greater than or...	60	PLATINUM
Else If Age	is greater than or...	45	GOLD
Else If Age	is greater than or...	25	SILVER

Add Clause

Else

BRONZE

OK

Cancel

Syntax: =IF(D2>=60,"Platinum", IF(D2>=40,"Gold", IF(D2>=25,"Silver", "Bronze")))

	ABC Gender	ABC City	ABC State	ABC Country	ABC 123 Loyalty_Level
55	Male	Donaldton	North Carolina	USA	GOLD
26	Female	Phillipsmouth	North Dakota	USA	SILVER
29	Female	Kylebury	Alabama	USA	SILVER
59	Female	Thomastown	Ohio	USA	GOLD
64	Female	Nicholasshire	California	USA	PLATINUM
24	Male	East Amandamouth	Oregon	USA	BRONZE
29	Female	Deborahborough	South Dakota	USA	SILVER
29	Female	West Pamelafurt	West Virginia	USA	SILVER
53	Female	Maynardton	New Hampshire	USA	GOLD
64	Male	Melissashire	Pennsylvania	USA	PLATINUM
53	Male	Port Michaelburgh	Pennsylvania	USA	GOLD
28	Female	East Krista	New York	USA	SILVER
55	Female	North Kathryn	Kansas	USA	GOLD
32	Female	New Becky	Montana	USA	SILVER
53	Female	Lake Kristamouth	Nebraska	USA	GOLD
49	Female	Lawsonhaven	Washington	USA	GOLD
24	Female	South Christopher	Pennsylvania	USA	BRONZE
60	Female	Brandonville	Georgia	USA	PLATINUM
43	Male	South William	Kansas	USA	SILVER
46	Female	Gonzalezfort	New Jersey	USA	GOLD
59	Male	Lake Shaneburgh	Delaware	USA	GOLD
36	Female	Smithburgh	Oklahoma	USA	SILVER
33	Female	Michaelton	Hawaii	USA	SILVER
52	Female	Molinabury	Oregon	USA	GOLD
40	Male	Rogerschester	Idaho	USA	SILVER
59	Female	Vasqueztown	California	USA	GOLD
18	Female	East Michellemouth	Missouri	USA	BRONZE
31	Male	Mollyshire	Utah	USA	SILVER
27	Female	New David	New Jersey	USA	SILVER
54	Female	East Jennifer	California	USA	GOLD
64	Female	Amybury	Pennsylvania	USA	PLATINUM
41	Male	Johnsonville	Oregon	USA	SILVER

Screenshot of Loyalty_Level Table:

Answer:

Loyalty_Level
GOLD
SILVER
SILVER
GOLD
PLATINUM
BRONZE
SILVER
SILVER
GOLD
PLATINUM

Syntax: =TEXT(Sales_Fact[Order_Date],"MMM")

Screenshot of MonthTable:

Answer:

J
Month
Jul
Dec
Jun
Nov
Jan
Jul
Jul
Nov
May
Jul
Feb
May
Aug
Aug
Jun
Dec
Sep

Syntax: =VLOOKUP(\$C2,Customer_Dim,5,FALSE)

Screenshot of Gender Table:

Answer:

Gender
Male
Female
Female
Female
Male
Female
Female
Female
Female
Female
Female
Female
Female

Syntax: =VLOOKUP(\$C2,Customer_Dim,9,)

Screenshot of Loyalty_Level Table:

Answer:

Loyalty_Level
GOLD
PLATINUM
GOLD
GOLD
SILVER
BRONZE
BRONZE
SILVER
PLATINUM
GOLD
BRONZE
SILVER
SILVER
GOLD
GOLD

Syntax: =VLOOKUP(\$D2,Product_Dim,3,FALSE)

Screenshot of Category Table:

Answer:

N	
Category	
Sports	
Electronics	
Beauty	
Electronics	
Beauty	
Electronics	
Home	
Home	
Home	
Home	
Sports	
Electronics	
Sports	
Electronics	
Sports	

Syntax: =VLOOKUP(\$D2,Product_Dim,4,FALSE)

Screenshot of Sub CategoryTable:

Answer:

Sub_Category
Team Sports
Audio
Fragrance
Computer
Haircare
Appliances
Decor
Storage
Kitchen
Decor

Syntax: =VLOOKUP(\$D2,Product_Dim,6,FALSE)

Screenshot of Cost Table:

Answer:

Cost
\$246
\$404
\$43
\$393
\$17
\$465
\$461
\$138
\$201
\$180
\$166
\$326
\$131
\$419
\$166
\$223

Syntax: =VLOOKUP(\$D2,Product_Dim,7,FALSE)

Screenshot of Stock Table:

Answer:

Stock
\$471
\$159
\$267
\$320
\$399
\$301
\$190
\$89
\$146
\$305
\$340
\$301
\$232

Syntax: =VLOOKUP(\$E2,Store_Dim,3,FALSE)

Screenshot of Region Table:

Answer:

Region
NORTH
NORTH
WEST
WEST
NORTH
SOUTH
SOUTH
WEST
NORTH
EAST
SOUTH
NORTH
SOUTH
WEST
NORTH
SOUTH

Syntax: : =VLOOKUP(\$E2,Store_Dim,5,FALSE)

Screenshot of Store_Type Table:

Answer:

S
Store_Type
OUTLET
FLAGSHIP
ONLINE
ONLINE
OUTLET
ONLINE
ONLINE
ONLINE
OUTLET
FLAGSHIP
OUTLET
FLAGSHIP
ONLINE
FLAGSHIP
FLAGSHIP
FLAGSHIP
ONLINE
ONLINE

Profit Margin

Syntax: $=[@[\text{Discounted_Revenue}]]-[@\text{Cost}]$

Screenshot of Profit Margin Table:

Answer:

Profit_Margin
-\$48
-\$176
\$2,699
\$381
\$1,816
-\$108
\$874
\$915
\$632
\$1,164
\$93
\$627
\$2,423
\$25
\$979
\$1,449
\$437
\$521
\$513

Syntax: =SUMIF(Sales_Fact[Category],"Sports",Sales_Fact[Total_Amount])

Answer: 635481

Syntax: =SUMIF(Sales_Fact[Gender],"Female",Sales_Fact[Total_Amount])

Answer: 1445317

Syntax: =AVERAGEIF(Sales_Fact[Category],"Sports",Sales_Fact[Profit_Margin])

Answer: 1112.9

Syntax: =XLOOKUP("Portable Power Bank",Product_Dim[Product_Name],Product_Dim[Stock])

Answer: 301

Syntax: =COUNTIF(Sales_Fact[Payment_Type],"PayPal")

Answer: 695

Syntax: `=XLOOKUP("Men's Hair Styling Gel",Product_Dim[Product_Name],Product_Dim[Stock])`

Answer: 399

❖ **Descriptive Analysis:**

In the Descriptive Analysis I used excel Formula and with using Toolpak.
Kindly refer the below syntax and Screenshots:

Total Amount

SUM–Syntax: `=SUM(J2:J2001)`

Mean(Average)–Syntax: `=AVERAGE(J2:J2001)`

Mode – Syntax: `=MODE(J2:J2001)`

Median– Syntax: `=MEDIAN(J2:J2001)`

Variance– Syntax: `=VAR.P(J2:J2001)`

SD– Syntax: `=STDEV.P(J2:J2001)`

Min– Syntax: `=MIN(J2:J2001)`

Max– Syntax: `=MAX(J2:J2001)`

Count– Syntax: `=COUNT(J2:J2001)`

Discounted Revenue

SUM–Syntax: `=SUM(K2:K2001)`

Mean(Average)–Syntax: `=AVERAGE(K2:K2001)`

Mode – Syntax: `=MODE(K2:K2001)`

Median– Syntax: `=MEDIAN(K2:K2001)`

Variance– Syntax: `=VAR.P(K2:K2001)`

SD– Syntax: `=STDEV.P(K2:K2001)`

Min– Syntax: `=MIN(K2:K2001)`

Max– Syntax: `=MAX(K2:K2001)`

Count– Syntax: **=COUNT(K2:K2001)**

Total_Amount		Discounted_Revenue	
Sum	\$25,54,771	Sum	\$21,72,989
Mean(Average)	\$1,277	Mean(Average)	\$1,086
Mode	136	Mode	24.49
Median	\$1,008	Median	\$869
Variance	901677.6389	Variance	669430.87
SD	949.5670797	SD	818.1875518
Min	\$21	Min	\$18
Max	\$3,992	Max	\$3,909
Count	2000	Count	2000

Summary Statistics using Toolpak

Decriptive Summary Total_Amount		Decriptive Summary Discounted_Revenue	
Mean	\$1,277	Mean	\$1,086
Standard Error	21.24	Standard Error	18.30
Median	\$1,008	Median	\$869
Mode	\$136	Mode	\$24
Standard Deviation	949.80	Standard Deviation	818.39218
Sample Variance	\$9,02,129	Sample Variance	\$6,69,766
Kurtosis	-0.13	Kurtosis	0.06
Skewness	0.82	Skewness	0.88
Range	\$3,971	Range	3890
Minimum	\$21	Minimum	\$18
Maximum	\$3,992	Maximum	\$3,909
Sum	\$25,54,771	Sum	\$21,72,989
Count	2000	Count	2000
Largest(1)	3992	Largest(1)	3909
Smallest(1)	21	Smallest(1)	18
Confidence Level(95.0%)	42	Confidence Level(95.0	36

1. Descriptive Analytics (What happened?)

- The dataset shows sales across multiple store types (flagship, outlet, online) and regions (North, East, South, West).

- Credit Card and PayPal dominate as payment methods, while COD is still common for mid-range purchases.
- Discounts range from 0% to 30%, with most transactions falling in the 5–15% range.

2. Diagnostic Analytics (Why did it happen?)

- Flagship stores generate large transactions because they target premium customers.
- Outlets attract bulk buyers due to higher discounts.
- Online stores show steady mid-range sales, reflecting convenience-driven purchases.
- High discounts greater than 25% reduce profitability, explaining why discounted revenue often drops sharply compared to total amount.

3. Predictive Analytics (What is likely to happen?)

- Flagship stores will likely continue to drive high-value sales, especially during festive seasons.
- Online sales will keep growing steadily as customers shift toward digital channels.
- COD usage may decline over time as digital payments gain trust.
- Discounts in the 5–15% range will remain the sweet spot for balancing sales volume and profitability.

4. Prescriptive Analytics (What should we do?)

- Maintain flagship stores as premium hubs but expand online presence for consistent revenue.
- Encourage digital payments with loyalty rewards to reduce COD reliance.
- Apply targeted discounts (5–15%) to maximize profitability while attracting customers.
- Use regional strategies: premium focus in North/East, digital-first in South/West.

Conclusion:

This study addressed the question of how inventory management affects sales performance in small retail businesses. The key findings revealed These patterns explain drivers of revenue differences across store types and discount levels. Future trends point toward digital growth and optimized discounting strategies. These actions guide decision-making to optimize sales and profitability.

